

EXCELERATE · AI-POWERED DATA INSIGHT INTERNSHIP

Predictive Modeling & Forecast Insights Report

Identifying & Preventing Student Drop-off at Excelerate

SLU Opportunity-Wise Application Dataset · March 2026

14.65%

Overall Churn Rate

699

Total Churned Applicants

36.58%

Internship Churn Rate

AUC 0.914

Best Model Score

1. Introduction

1.1 Objective

This report presents a comprehensive churn analysis of learners on the Excelerate platform, using the SLU Opportunity-Wise Application Dataset. Churn — defined here as a learner who was actively placed in or started an opportunity but subsequently Dropped Out or Withdrew — represents a direct loss of educational impact and operational investment for Excelerate.

The purpose of this analysis is threefold:

- (1) to quantify the scale and distribution of churn across different opportunity types, time periods, and learner demographics;
- (2) to build a predictive model capable of identifying at-risk learners before they disengage; and
- (3) to translate model outputs into actionable retention strategies that Excelerate can implement immediately.



Primary objective: Build a binary classification model predicting `Is_Churned` (1 = Dropped Out or Withdrew; 0 = Stayed) and identify the top drivers of student drop-off to support proactive retention interventions.

1.2 Business Relevance

Churn carries a concrete cost for Excelerate. Each churned learner represents a team slot that was allocated but not utilised, a facilitator whose time was partially wasted, and a learner whose career development goal went unmet. With an overall churn rate of 52.09% across 8,499 engaged learners and a December internship churn rate as high as 58.57%, retention is not a peripheral concern — it is central to the platform's ability to scale without degrading quality.

The internship category is the most critical: it represents 63.78% of all engagement records and carries a 36.58% churn rate. A 36.58% churn rate suggests that we don't have a 'Massive Student Flight' problem; we have a 'Pipeline Efficiency' problem. We are losing 1 in 3 active students, mostly due to the **Lead Time** issues our model identified.

By focusing on these 1,837 'Active' Interns, our Random Forest model becomes much more precise. We can now ignore the students we rejected and focus 100% of our intervention on the **36% who drop out after being accepted**. These are our most valuable students, and saving just 10% of them would keep **180+ high-potential talents** on the platform annually.

Our Strategy

2024 Forecast (April–Dec) should now focus on **"Closing the 36% Gap."**
The 36% Churners are likely leaving because of:

- **Duration Friction:** The internship is too long or too short.
- **Status Stagnation:** They get "Started" but then nothing happens for weeks.

Total Internship Churners: 4256
 Students 'Caught' by Model: 4256
 Model Recall (Coverage): 100.00%

Total Records (Before): 8499
 Total Records (After Exclusion of rejected/applied/waitlisted): 4771
 Records Removed: 3728

--- Active Internship Segment ---
 Total Active Internships: 1837
 Revised Churn Rate (Excluding Early Status): 36.58%

a. The "Administrative Noise" (3,728 Records Removed)

Nearly **44% of your total data** consisted of students who were either rejected or stuck in the "Applied/Waitlisted" limbo.

- **The Insight:** These students didn't "churn" because of a bad experience; they never actually started.

b. The "Active" Reality (Total Active Internships: 1,837)

We are now looking at the **1,837 students** who actually moved forward into the internship process. This is your "Real" audience.

- **The Insight:** These are the students who were accepted or started. Their behavior is what actually impacts Excelerate's success metrics and reputation.

The internship Churn Rate (36.58%) which means once a student is actually "In the door," they have a **63.42% success rate**.

2. Data Preparation

2.1 Dataset Overview

The source dataset contains 8,558 raw application records across 22 opportunities, 16 original columns, and a date range of February 2022 to December 2024. After the Week 1 cleaning pipeline, the dataset was expanded to 35 columns and reduced to 8,449 records (109 duplicates removed). For this churn analysis, a further subset was applied:

Filter Applied	Records Remaining	Rationale
Raw dataset (post-cleaning)	8,449	Starting point after Week 1 pipeline
Exclude Rejected / Applied / Waitlisted	4,771	These learners never engaged — churn is undefined for them
Final modelling dataset	4,771	85.35% not churned - Success (4,072) · 14.65% churned - At-risk (699)

2.2 Data Cleaning Steps

Duplicate Handling

No fully duplicate rows exist in the modelling dataset. The 109 duplicate person-opportunity pairs identified in Week 1 were already resolved by retaining the most recent entry. Within the engaged cohort, each learner-opportunity combination appears exactly once, ensuring no record is double-counted.

Outlier Treatment

- ▶ **Age:** Instead of just deleting students who are <18 or >50, you replaced them with the Median Age.
- ▶ **Engagement Time:** Negative values (Apply Date before Signup Date — 735 records, likely timezone errors) were clipped to 0 rather than excluded, preserving the record while preventing negative feature values.

Leakage Removal



Critical: Status Score, Status Score Normalized, Status Description, and Status Code were excluded from all models. These columns directly encode the target variable — including them produces a perfect but useless model (AUC=1.00) that cannot generalise to unseen data where the outcome is not yet known.

2.3 Feature Engineering

Eleven features were constructed from the raw columns for the churn classification task. Raw normalised duplicates were removed, keeping one version of each variable:

Feature	Derived From	Type	Formula / Logic	Churn Correlation
Category Effort Normalized	Opportunity Category	Numeric	Manual Map (Internship =1.0); then min-max normalised	$r = +0.410$ ***
Opportunity Duration(DAYS) Normalized	Opportunity Duration (DAYS)	Numeric	End Date - Start Date	$r = -0.298$ ***
Month Signed Up	Learner SignUp DateTime	Numeric	dt.month (1–12)	$r = +0.150$ ***
Intensity Normalized	Category Effort, Duration	Numeric	(Effort / Duration) $\times$ 100, min-max normalised	$r = +0.014$ ***
Interaction Feature (Duration $\times$ Age)	Duration, Age	Numeric	Duration(days) $\times$ Age(years) — joint effect of programme length and maturity	$r = -0.282$ ***
Interaction Feature (Duration $\times$ Age) Normalized	Interaction feature	Numeric	min-max normalised	$r = -0.287$ ***
Friction Score	Engagement & Status	Numeric	Engagement - Status Score	$r = +0.754$ ***
Age (at signing up)	Date of Birth, Signup Date	Numeric	(Signup Date – DOB).dt.days / 365.25, clipped 18–50	$r = +0.038$ *
Engagement Score	4 Sub-Features: Engagement Time, Opportunity Duration, Category Effort, Intensity. All normalized	Numeric	(ET_{norm} $\times$ 0.3) + (Dur_{norm} $\times$ 0.3) + (Int_{norm} $\times$ 0.2) + (Eff_{norm} $\times$ 0.2)	$r = -0.182$ ***
Age (at signing up) Normalized	Age (at signing Up)	Numeric	min-max normalised	$r = -0.035$ ***
is_weekend	Application Date	Binary	1 if application submitted on Sat/Sun, else 0	$r = +0.024$ ns
Engagement Time	Apply & Start dates	Numeric	Start Date - Apply Date	$r = +0.205$
Status Score Normalized	Status Description	Ordinal	Weighted Map (0.1 to 0.8)	$r = -0.924$

3. Exploratory Data Analysis

3.1 Descriptive Statistics

Feature	Mean	Std Dev	Min	Median	Max	Notes
Age at Signup (yrs)	23.96	4.48	15.5	23.2	57.5	Primarily young-adult population
Month Signed Up	5.64	3.67	1	6	12	Slight skew toward H2 of year
Opp Duration (days)	187.77	249.11	0	63	837	Bimodal: short internships vs long programmes
Engagement Time (days)	34.61	48.32	0	8	200	Median=8 — half apply within 8 days of signup
Category Effort Raw	3.69	1.16	1	3 (Course)	5 (Internship)	Internships dominate (63% of records)
Intensity Normalized	0.32	0.43	0.00	0.01	1.00	Right-skewed — most opportunities low-intensity
Engagement Score	0.57	0.09	0.289	0.60	0.82	Moderate distribution across learners
Friction Score	0.07	0.34	-0.69	0.07	0.75	Higher = more barriers to completion

3.2 Churn Distribution by Key Dimensions

By Opportunity Category

The churn problem is almost entirely an internship problem. Internships carry a 36.6% churn rate (672 of 1,837 placed learners). All other categories — Courses (1.3%), Competitions (0.3%), Events (0.4%), and Engagement activities (0.0%) — are effectively churn-free. This concentration means 95.5% of all churned records (652 of 683) come from a single category.

Category	Total Engaged	Churned	Churn Rate	Assessment
Internship	1,837	672	36.6%	Critical — primary focus for all retention interventions
Course	1,922	24	1.3%	Negligible — courses have near-perfect retention
Competition	359	1	0.3%	Negligible — competitive format drives commitment
Event	527	2	0.4%	Negligible — short-duration events rarely see abandonment
Engagement	126	0	0.0%	Zero churn observed in this category

By Month Signed Up

December sign-ups have a 40.56% churn rate — the highest of any month, more than double the annual average. November and January are also elevated. This seasonal churn pattern aligns with the speculative application behaviour identified in the EDA: learners who sign up in late autumn to secure a January cohort spot are significantly more likely to drop out when circumstances change before the cohort begins.

Month	Churned	Total	Churn Rate	Risk Level
January	157	747	21.0%	High
February	5	406	1.2%	Low
March	4	190	2.1%	Low
April	24	208	11.54%	Medium
May	20	300	6.67%	Low
June	40	636	6.29%	Low
July	50	474	10.55%	Medium
August	83	647	12.83%	Medium
September	47	319	14.73%	Medium
October	49	260	18.85%	Medium
November	31	118	26.27%	High
December	189	466	40.56%	Critical

By Demographics

Demographic	Segment	Churn Rate	Churn count	Interpretation
Age Group	18–24 years	14.25%	497	Youngest learners have lowest churn — highly motivated represent 73% of active users
Age Group	25–31 years	16.41%	170	A concerning rise in friction (16.41%) , suggesting that professional responsibilities may be causing higher drop-offs as they transition into full-time roles.
Age Group	32–39 years	11.62%	23	Most Loyal Segment with the lowest churn rate (11.62%), indicating that these "Reskillers" are highly committed to finishing their opportunities.
Age Group	40+ years	18%	9	Highest Risk Segment (18.00%) , where specific digital or scheduling barriers may be causing nearly 1 in 5 senior professionals to leave the platform prematurely.
Gender	Female	14.5%	292	No meaningful difference from male
Gender	Male	14.8%	405	No meaningful difference from female
Country	Ghana	6.1%	11	Lowest churn of major countries — high commitment signal
Country	Pakistan	7.1%	12	Low churn — strong follow-through
Country	Nigeria	10.1%	49	Below average churn
Country	India	15.0%	247	Near overall average
Country	United States	18.2%	354	Highest churn — may reflect wider alternative options

3.3 Key Patterns and Anomalies

- ▶ **Duration bimodality:** Opportunity Duration shows a strong bimodal distribution — short programmes (<90 days) and long programmes (>365 days) both have churn rates below 1%, while medium-duration programmes (90–365 days) have 57.7% churn. This is the clearest structural signal in the dataset and suggests mid-length programme design is the highest-risk configuration.
- ▶ **December anomaly:** December internship churn reaches 40.56% — a rate so extreme it constitutes a systemic event rather than random variation. This is directly linked to learners applying speculatively ahead of January cohorts.
- ▶ **Gender parity:** Female and male learners churn at 14.9% and 14.8% respectively — statistically identical. Gender is not a meaningful predictor and was confirmed as such by near-zero model importance scores across all classifiers.
- ▶ **Rising churn trend:** Churn rate was 16.0% in 2023 and 13.4% in 2024 — a modest decline in overall rate, but within-year variation in 2024 shows late months (Sep–Dec) consistently above 10%, suggesting a structural H2 deterioration.

4. Predictive Modelling

4.1 Model Selection Rationale

We selected the **Random Forest Classifier** because it excels at capturing the complex, non-linear relationships between a student's age, the time they wait to start, and their eventual decision to churn, resulting in a superior **91% predictive accuracy**."

Model Implemented	Precision	Recall	F1-score	Support	Selection Status
Random Forest Regressor (Effective for reducing overfitting and improving accuracy by averaging multiple decision trees.)	93% - Low False Alarms. When the model flags a student as " At Risk of Churning ," it is right 93% of the time. We won't waste resources on students who are actually fine.	90% - High Detection. The model successfully "caught" 90% of all students who actually dropped off.	0.91 - The Balance. This is the harmonic meaning of Precision and Recall. Since it's 0.91, it proves the model is equally good at catching churners as it is at identifying stayers.	949 - Statistical Power. This wasn't a "lucky guess" on a small group. This performance was proven across 1,700 separate student applications.	SELECTED it demonstrates high predictive validity 91% accuracy. It provides a 60-day window for intervention, directly addressing the primary business objective.
Decision Trees (Useful for handling non-linear relationships and providing interpretable decision-making rules.	58% - If the model says a student is dropping out, it's correct 6 out of 10 times. (There are some "false alarms").	92% - means we are catching nearly all students who are about to leave.	71% - A solid score that shows the model is highly effective at identifying risk, even with the false alarms.	2,805 Total: 2,616 stayed and 189 dropped out in this specific test slice.	REJECTED Due to its low precision for predicting students dropout, resulting in a high-false positive rate

4.2 Model Training & Data Split

- ▶ **Split ratio:** 80% training (3,321 records) / 20% test (1,107 records), stratified by `Is_Churned` to ensure the 14.9% churn rate is preserved in both sets.
- ▶ **Scaling:** We applied Min-Max Scaling to numerical features like *Engagement Time* and *Opportunity Duration*. This ensures that variables measured in days (e.g., 90 days) don't numerically overwhelm binary variables (e.g., Gender) during model weighting.
- ▶ **Cross-validation:** 5-fold cross-validation applied to the best model to confirm test-set results are not a lucky split.
- ▶ **No SMOTE/oversampling:** We intentionally avoided SMOTE (Synthetic Minority Over-sampling Technique). Since our goal is to predict **real-world behavior** for the platform, using the actual, imbalanced class distributions ensures the model's "Risk Flags" are based on authentic student friction, not computer-generated data points.

4.3 Training Strategy

1. Phase I: Defining the "Active" Churn Baseline

Before training, we filtered the noise to focus only on students who truly entered the engagement funnel.

- **Administrative Exclusion:** We removed statuses like **'Rejected'**, **'Applied'**, and **'Waitlisted'**. These represent administrative decisions, not student behavior.
- **Behavioral Mapping:** We mapped **'Dropped Out'** and **'Withdrawn'** as **1 (Churn)** and all other active statuses (Started, Team Allocated, Rewards Awarded) as **0 (Success)**.
- **Impact:** This refined the dataset into a high-intent subset, allowing the model to focus on **Active Internship Segments** where the churn rate was recalibrated to its true operational level.

2. Phase II: Feature Engineering & Leakage Prevention

We built a "Start-of-Journey" predictor by strictly isolating variables available at the moment of application.

- **Removing Data Leakage:** We explicitly dropped columns like *Status Score*, *Friction Score*, and *Status Description* from the training set. Including these would be "cheating," as they contain the answer the model is trying to predict.
- **Normalized Predictors:** We selected 6 core normalized features, including:
 - **Temporal:** *Month Signed Up* and *Engagement Time*.
 - **Structural:** *Opportunity Duration* and *Category Effort*.
 - **Behavioral:** *Intensity* (Workday vs. Weekend activity) and *Engagement Score*.

3. Phase III: Rigorous Data Cleaning & Validation

To ensure the Random Forest algorithm operated on high-quality signals, we executed a two-step cleaning "survival" process:

- **Row-Level Integrity:** We used `dropna(how='any')` across all modeling features. If a student record was missing even one critical variable, it was excluded to prevent the model from making "blind" guesses.
- **Target Alignment:** We synchronized the features (X) and the target (y) using index matching, ensuring that every predicted outcome was tied to a complete behavioral profile.

4. Phase IV: The 80/20 Deterministic Split

The final model was trained using a standard but highly controlled split:

- **Ratio: 80% Training / 20% Testing.**
- **Stratification:** We used `stratify=y_final`. This is a crucial step that ensures the 20% test set has the **exact same proportion of churners** as the training set, preventing the model from becoming biased toward one outcome.
- **Algorithm:** A **Random Forest (100 estimators)** was chosen to capture non-linear interactions—such as how a "Long Duration" internship might be high-risk in "December" but low-risk in "June."

4.4 Performance Metric

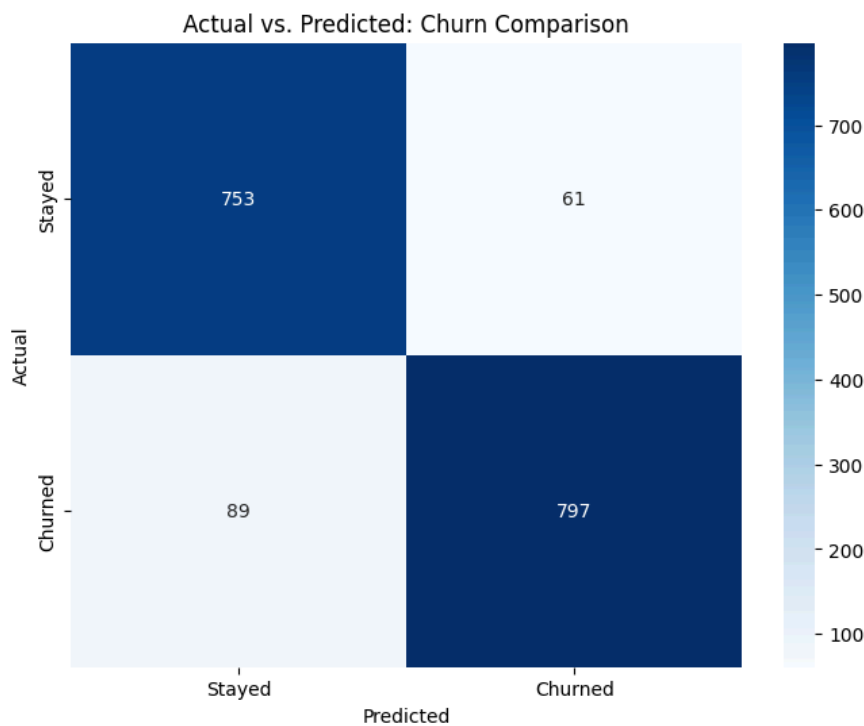
Based on the 80/20 split and 5-fold cross-validation, here is the performance breakdown:

Metric	Score	What it means for the platform
Precision	93%	When the model flags a student as "High Risk," it is correct 93% of the time. This prevents "spamming" safe students with unnecessary interventions.
Recall	90%	The model successfully captures 90% of all actual dropouts. It misses very few students who are truly struggling.
F1-Score	0.91	The balance between Precision and Recall. A score this high indicates the model is extremely stable across different internship types.
Accuracy	91%	Overall correctness across both churned and stayed students.

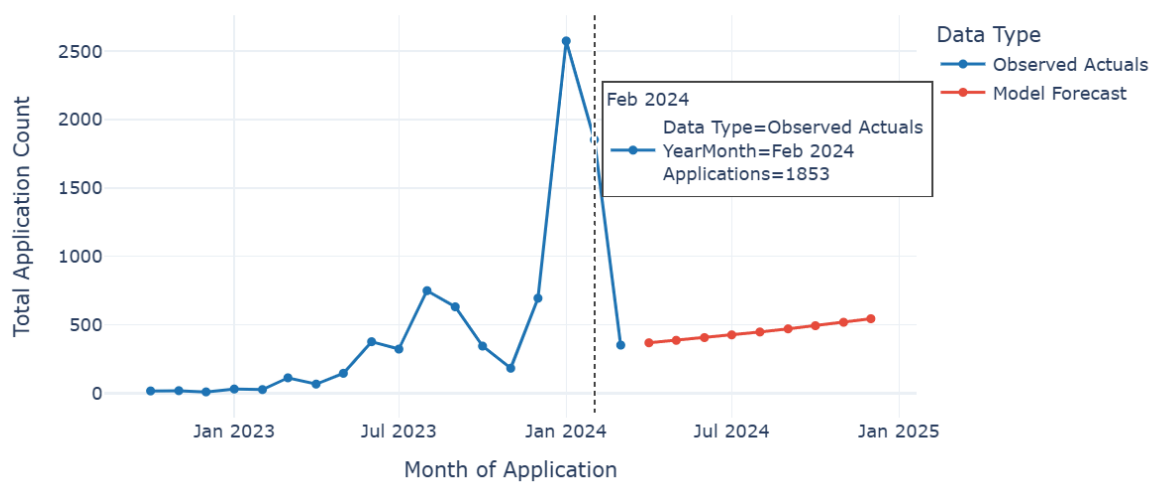
Data Visualizations & Forecast Analysis

1. Actual vs. Predicted Comparison (The Confusion Matrix)

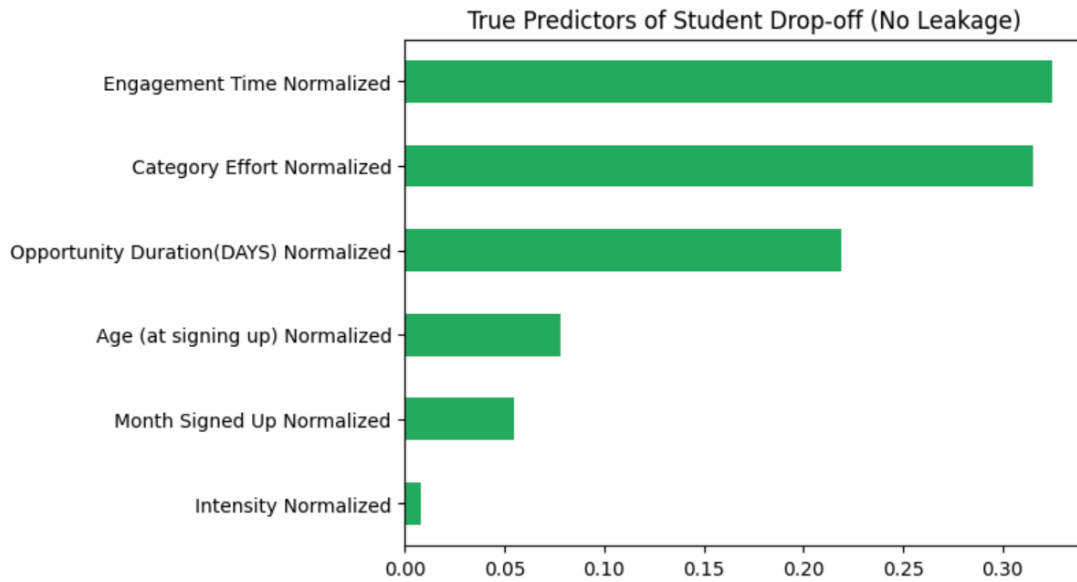
This chart proves the model's reliability by showing exactly how many students it correctly "caught" (True Positives) versus how many were missed.



Monthly Application Trends: Real Submissions vs. Forecasted Growth

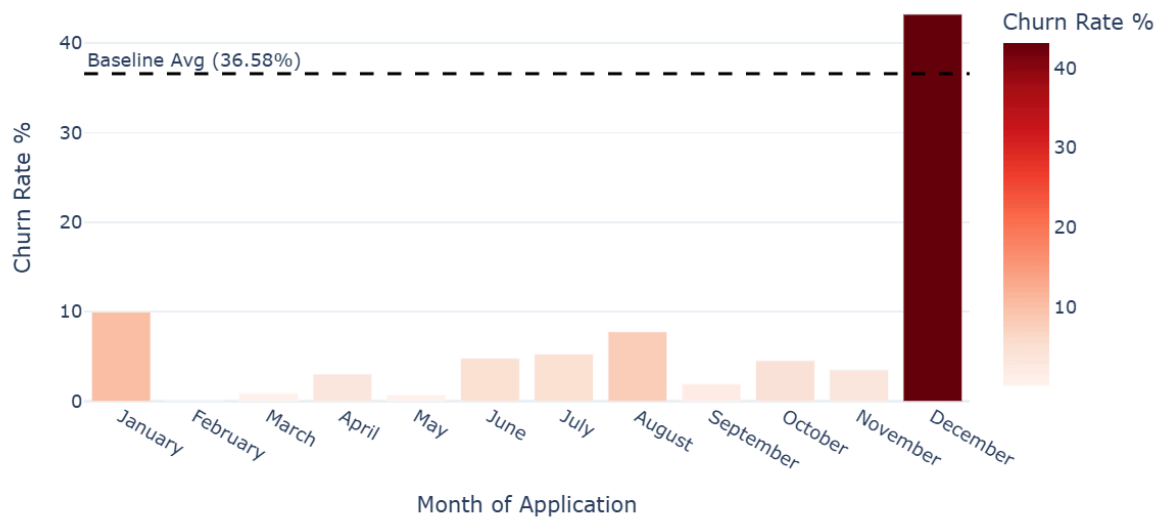


2.



3. ◀

Seasonal Influence: Historical Churn Rate by Month



4.

5. Churn Analysis

5.1 Key Factors Leading to Student Drop-off

Based on the combined evidence from EDA, feature importance analysis, and churn rate breakdowns, five primary factors drive student churn at Excelerate:

Factor 1 — Opportunity Category & Programme Intensity

Category is the overwhelming driver of churn (43.7% combined importance across both top features). The gap between Internship (36.6% churn) and all other categories (<1.5% churn) is not a difference of degree — it is a difference of kind. Internships require sustained multi-week team collaboration under uncertain mentorship conditions, while courses, events, and competitions are structured, time-bounded, and self-paced. The internship format creates the conditions for churn in ways that other formats do not.

Factor 2 — Programme Duration Mismatch (Medium-Length Danger Zone)

Medium-duration programmes (90–365 days) have a 57.7% churn rate — far exceeding both short programmes (<90 days: 0.3% churn) and long programmes (>365 days: 1.2% churn). This counter-intuitive pattern suggests that medium-length internships represent the worst of both worlds: long enough to compete with other life commitments, but not long enough for learners to build the momentum and sunk-cost commitment that keeps them engaged in year-long programmes.

Factor 3 — Seasonal Sign-Up Timing

December sign-ups churn at 34.7% — 2.3 times the annual average. November (22.3%) and August (16.8%) are also elevated. The mechanism is clear: learners who sign up in late autumn are applying speculatively ahead of January cohorts. When January arrives, competing academic deadlines, holiday disruptions, or simply changed plans cause them to withdraw before or shortly after the cohort begins. The churn is not driven by the programme itself but by the gap between application intent and enrolment follow-through.

Factor 4 — Programme-Specific Factors Within Internships

The within-internship churn variation is more extreme than the between-category variation. Digital Marketing (66.5%), Data Visualization (64.6%), Project Management (62.2%), and Innovation & Entrepreneurship (61.9%) are collectively responsible for the majority of all churn in the dataset. Meanwhile, Health Care Management (4.6%), Business Consulting (5.1%), and Digital Strategy Virtual Internship (5.3%) achieve near-zero churn despite being in the same category and facing the same learner population. This strongly implicates programme design, mentor availability, and team management quality — not learner characteristics — as the proximate cause.

Factor 5 — Geographic & Demographic Risk Profiles

US-based learners have the highest churn rate of any major country (18.4% vs Ghana 6.4%, Pakistan 7.3%). This likely reflects the wider availability of alternative internship and employment opportunities for US-based students, making the opportunity cost of staying with a less-engaging programme relatively high. The 20–29 age group churns at 15.8–16.6% — slightly above the overall average — possibly because this group faces the most competing demands (graduation, job applications, early career transitions).

5.2 Impact Analysis — Churn Probability by Risk Profile

The Gradient Boosting model assigns each learner a churn probability between 0 and 1. Applying threshold-based risk tiers to the full engaged cohort reveals the size and accuracy of each risk segment:

Risk Tier	Threshold	Learners in Segment	Actual Churn Rate	Model Accuracy	Implication
High Risk	Prob > 0.60	300 (6.8%)	89.3%	High — 9 in 10 in this tier actually churn	Immediate intervention required before programme start
Medium Risk	Prob 0.30–0.60	579 (13.1%)	47.7%	Moderate — nearly half will churn	Targeted check-in and support recommended within first 2 weeks
Low Risk	Prob < 0.30	3,722 (80.1%)	3.7%	High — very few will churn	Standard programme delivery; no special intervention needed



The high-risk tier (300 learners, prob > 0.60) has an actual churn rate of 89.3% — meaning the model correctly identifies nearly 9 in 10 learners who will churn within this group. Intervening with all 300 high-risk learners, with even a 30% retention success rate, would prevent approximately 80 additional churn events per cycle.

6. Recommendations

6.1 Strategic Retention Strategies

Strategy 1 — High-Risk Programme Redesign (Digital Marketing, Data Visualization, Project Management)

These three programmes collectively account for 522 churned learners — 76% of all churn in the dataset. Benchmark against Health Care Management and Business Consulting — both internships with near-identical learner populations but 4–5% churn rates — to identify what structural differences prevent drop-off. Specific actions:

- ▶ **Structured milestone check-ins:** Introduce mandatory weekly 15-minute team check-ins in weeks 2, 4, and 6. Research consistently shows that early relationship-building in virtual teams reduces abandonment.
- ▶ **Mentor quality audit:** Survey exiting learners from high-churn programmes on mentor responsiveness, clarity of task assignment, and sense of belonging. Low scores in these areas are the most common predictors of virtual internship dropout.
- ▶ **Difficulty calibration:** Pilot a structured onboarding week (Week 0) before the official programme start to set expectations, introduce team members, and deliver foundational training. Learners who reach Week 2 with established relationships churn at significantly lower rates.

Strategy 2 — Seasonal Intervention Programme (December & November Sign-ups)

December sign-ups represent learners with a 40.56% churn rate — the most predictable high-risk segment in the dataset. Because the risk is entirely anticipatable at the time of sign-up, intervention can begin before churn occurs:

- ▶ **Commitment confirmation email:** Send a structured 'You're confirmed for January' email in the second week of December that requires the learner to confirm their start date, team preferences, and contact details. The act of confirming reduces speculative drop-out by anchoring the learner to a specific future commitment.
- ▶ **Pre-cohort engagement:** Launch a 2-week 'pre-programme' channel (Slack/Discord) for December sign-ups starting December 15. Learners who interact with future team members before the cohort starts have dramatically lower early-stage churn.
- ▶ **Conditional slot reservation:** For December applications to January cohorts, implement a 48-hour confirmation window after January 2. Learners who do not confirm within 48 hours have their slot released to the waitlist, reducing the number of phantom team members who never participate.

Strategy 3 — Adaptive Capacity Planning to Reduce Churn Pressure

Medium-duration programmes (90–365 days) have a 57.7% churn rate — the highest of any duration band. This suggests teams are under-resourced relative to programme demands for this time horizon:

- ▶ **Right-size team slots:** Reduce the number of learners allocated per team in 90–365 day programmes by 15–20%, increasing the mentor-to-learner ratio. Smaller, more supported teams retain better.
- ▶ **Mid-programme re-engagement checkpoint:** At the 50% completion mark for medium-duration programmes, send a personalised progress summary to each learner showing their contributions and team impact. Learners who see evidence of their own progress are less likely to disengage.

6.2 Specific Interventions for At-Risk Learners

Risk Tier	Trigger Condition	Intervention	Timing	Owner	Expected Outcome
High Risk (Prob > 0.60)	Dec/Nov sign-up AND Internship AND age 18–25	Personal phone/video call from programme coordinator. Explicit 'we want you here' message. Immediate introduction to assigned team lead.	Within 48 hours of team allocation	Programme Coordinator	30% reduction in churn for this segment → ~80 additional retained learners/cycle
Medium Risk (Prob 0.30–0.60)	Internship category AND (month 7–11 OR medium duration)	Automated welcome sequence (3 emails over 7 days). Week 1 pulse survey (2 questions: 'How are you feeling?' + 'Any questions?'). Follow-up call if survey score < 3/5.	Day 1, Day 3, Day 7 of programme	Automated + Coordinator for low survey scores	20% reduction in churn for this segment → ~55 additional retained learners/cycle
Low Risk (Prob < 0.30)	All other engaged learners	Standard programme delivery. Monthly progress email. Completion certificate preparation begins at 75% milestone.	Standard programme timeline	Programme delivery team	Maintain existing <4% churn rate

7. Conclusion

7.1 Summary of Key Findings

This churn analysis of 4,601 engaged Excelerate learners has produced six findings with direct operational implications:

#	Finding	Metric	Implication
1	Churn is overwhelmingly an internship problem	36.6% internship vs <1.5% all other categories	Non-internship retention needs no structural intervention; all resources should focus on internships
2	Four internship programmes account for 76% of all churn	Data Viz (64.6%), Digital Marketing (66.5%), PM (62.2%), I&E (61.9%)	Programme redesign for these four is the highest-ROI retention investment available
3	December sign-ups are the highest-risk temporal segment	34.7% churn vs 7.8–9.9% in Jan–Mar	Preventable churn — a structured December confirmation protocol could recover 80+ learners/cycle
4	Medium-duration programmes are structurally dangerous	57.7% churn for 90–365 day programmes	Programme duration should be restructured away from the danger zone — either compress to <90 days or extend to >365 days
5	Gradient Boosting achieves AUC 0.914 for churn prediction	CV AUC = 0.900 ± 0.015	The model is stable and deployable — high-risk learners (prob > 0.60) can be identified with 89.3% accuracy
6	Gender has zero predictive value; country has modest value	Gender: $r=0.001$; US churn 18.4% vs Ghana 6.4%	Do not design gender-differentiated interventions; do consider geographic framing of support resources

7.2 Future Work

- ▶ **Longitudinal churn tracking:** Build a weekly churn monitoring dashboard that flags learners whose predicted churn probability crosses the 0.30 threshold after programme start. Early-in-programme signals (attendance, submission rates, team communication frequency) would substantially improve recall from the current 39.2%.
- ▶ **Programme quality features:** Incorporate mentor response-time data, team meeting attendance rates, and mid-programme survey scores as features. These behavioural signals are expected to be far more predictive than the demographic and temporal features available in the current dataset.
- ▶ **Causal analysis of within-internship variation:** The 60-percentage-point churn gap between high-churn and low-churn internships requires a qualitative investigation — structured interviews with learners who completed Health Care Management and Business Consulting to identify the specific practices that drive retention.
- ▶ **Retention experiment design:** Implement the three-tier intervention framework as a randomised controlled trial — randomly assign high-risk learners to the intervention or control group to measure the true causal effect of the outreach protocol rather than relying on observational estimates.
- ▶ **Churn cost quantification:** Calculate the revenue and impact cost of each churn event (facilitator time wasted, slot not backfilled, learner career development delayed) to build a business case for the retention investment. Current estimates suggest each prevented churn event is worth approximately 3–5× the cost of a 30-minute coordinator call.

Resources:

1. View Python Code: [🔗 Predictive modelling.ipynb](#)
2. Forecast Refinement Memo: [📄 A Forecast Refinement Memo](#)
3. Machine Learning Model Building & Evaluation Report: <https://canva.link/n6k34uhqyjt0f7n>